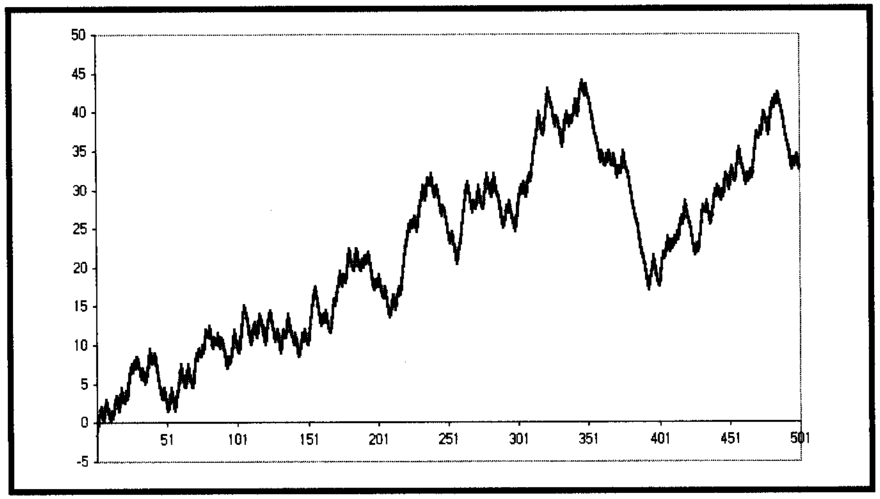

FIGURE 15.2 Another Hypothetical Equity Growth Example for %Profitable = 45 and Profit Factor = 1.50

equity curves of some of our systems at www.mesa-systems.com. Next, explore what the lower-limit statistics might be for a profitable trading system. My experience is that the boundary is 42 in cell A2 (for percentage winners) and 1.5 (for Profit Factor) in cell B2.

KEY POINTS TO REMEMBER

- Profit Factor and Percentage Winners of a trading system are all you need to create a Monte Carlo equity curve of that system.
- A real equity curve is only one of the possibilities that can be produced by a Monte Carlo equity curve.
- A Monte Carlo simulation can be used to evaluate the expected performance of any trading system.

CHAPTER 16

Leading Indicators

"Leading indicators are neat," said Tom predictably.

There are two basic kinds of leading indicators: causal and noncausal filters. Causal filters depend on data and noncausal filters can be predictive from almost any other basis, including gut feelings. The Sinewave Indicator described in Chapter 11 is an example of a noncausal filter. The purpose of this chapter is to derive the limitations and usefulness of causal predictive filters. It is a fundamental principle that causal filters cannot predict a specific event because their very value depends on that event. That is to say, causal filters cannot anticipate a transient response. However, they can and do act as reliable indicators of steady-state responses.

All moving averages have lag. A moving average is depicted as the dashed line relative to the original function (the solid line) in Figure 16.1a. The difference between the two lines d is a constant value in the case of a continuous trend. Similarly, the lag k is also a constant value. The leading indicator is created by adding the difference between the original function and its moving average to the function itself. Adding the difference necessarily places the indicator with a negative lag relative to the original function, as depicted in Figure 16.1b. Negative lag makes this filter a leading indicator. The amount of lead is exactly equal to the amount of lag of the moving average.

Since the amount of lead of the leading indicator is dependent on the lag of a moving average, it is instructive to examine the lag of an exponential moving average as a function of its smoothing parameter alpha. Imagine an original function that increases by 1 with each sample. The function will have a value of I on the Ith sample. If the moving average has a lag of k, then the moving average will have a value of $(I - k)$ on the Ith day.

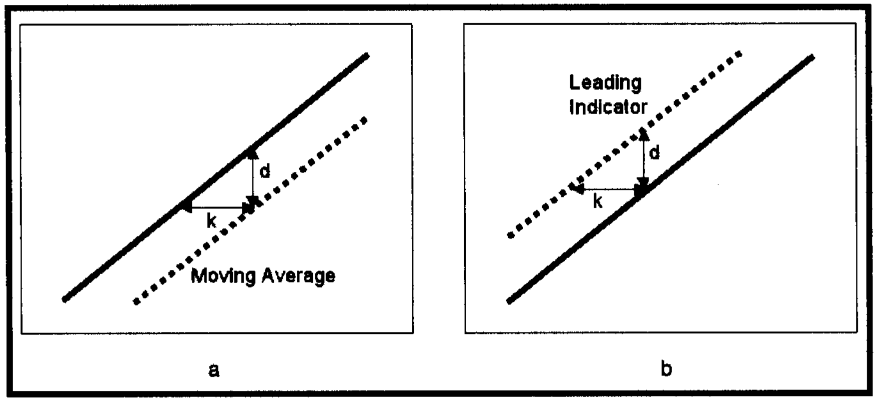

FIGURE 16.1 How Leading Indicators Are Constructed
a. A moving average has a lag *k* and a difference *d*.
b. Adding the moving average difference yields a lead *k*.

Similarly, the moving average will have had a value of $(I - 1 - k)$ on the $(I - 1)$th day. Putting these values in the equation for an exponential moving average, we have

$$(I - k) = \alpha * I + (1 - \alpha) * (I - 1 - k) \tag{16.1}$$

Solving for alpha in terms of the delay k, we have the relationship

$$\alpha = 1/(k + 1) \tag{16.2}$$

Or, conversely

$$k = 1/\alpha - 1 \tag{16.3}$$

Equation 16.3 tells how much lead we can expect from our leading indicator. From Chapter 2, the transfer response is the ratio of the output to the input. Thus the transfer response of the leading indicator can be written as

$$\begin{aligned} H(z) = \frac{\text{Output}}{\text{Input}} &= 2 - \frac{\alpha * Z^{-1}}{1 - (1 - \alpha) * Z^{-1}} \\ &= \frac{2 + (\alpha - 2) * Z^{-1}}{1 - (1 - \alpha) * Z^{-1}} \end{aligned} \tag{16.4}$$

But there is a price to be paid for getting the leading function. That price is noise gain. If we let $Z^{-1} = 1$ in Equation 16.4, we get the zero frequency

(constant input) gain. Doing this algebra, the gain of this filter is unity. That is, if the input is constant we get exactly the same output from the filter. The output cannot be leading because there is no trend to the input. Letting $Z^{-1} = -1$, the value of the transfer response at the Nyquist (highest possible) frequency is obtained. Doing this, the filter gain for a two-bar cycle is $(4 - \alpha)/(2 - \alpha)$. So the noise gain varies from 2 when $\alpha = 0$ to 3 when $\alpha = 1$. If the lead is three bars, Equation 16.2 gives $\alpha = 0.25$, and therefore the noise gain is 2.14, slightly more than 6 dB. Figure 16.2 shows how the noise gain varies with frequency for the case when $\alpha = 0.25$.

Noise gain is not a good thing. The noise gain can be reduced by following the leading indicator filter with an exponential moving average. As I indicated earlier, all moving averages have lag. So, if an alpha of the moving average is selected to have less lag than the lead of the leading indicator, an indicator having a net leading function can still be produced. As an example, selecting $\alpha = 0.33$ results in an exponential moving average that has a lag of only two bars. The attenuation at $Z^{-1} = 1$ is 0.2, which gives a greater attenuation than the noise gain of the leading indicator. The net gain of the composite filter is shown in Figure 16.3. While there is still some noise gain in the vicinity of a 20-bar cycle (frequency = 0.05), the net filter has a net smoothing effect over most of the frequency range.

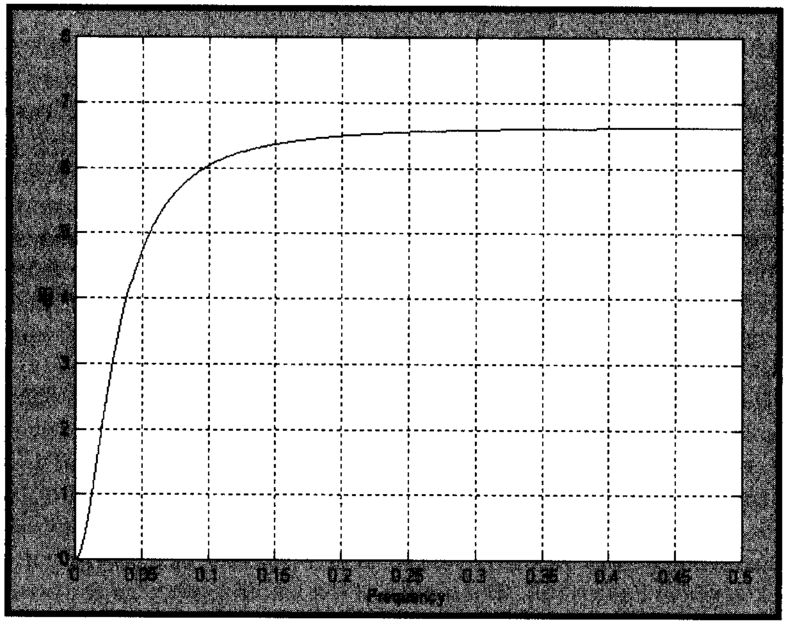

FIGURE 16.2 Noise Gain of a Leading Indicator

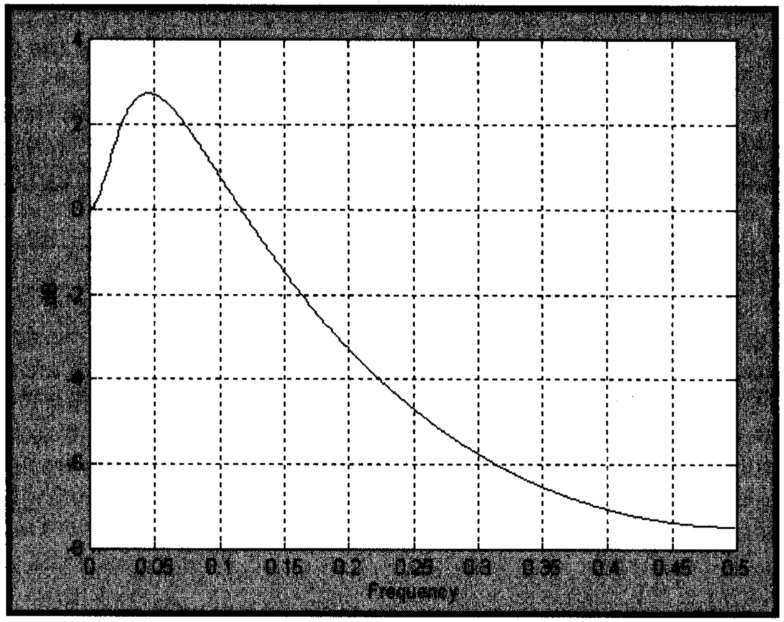

FIGURE 16.3 Net Gain of a Leading Indicator

The leading characteristic is still present in the net filter, as shown in Figure 16.4. As predicted, the lead is one bar at very low frequencies. That is, the trend indication will lead by one bar. However, the net filter has a lag of approximately 2.5 bars for cycle components near 20-bar cycles. Also, higher-frequency lag settles down to be about half a bar. The interpretation of the lag response is that the filter predicts a continuation of a trend by 1 bar, lags abrupt changes by about 0.5 bars, and lags smooth changes that can be fitted by segments of a 20-bar sinewave by as much as 2.5 bars. That's the law of physics—you cannot get something for nothing. Causal filters can have a predictive capability over some portion of the frequency response, but not at all frequencies. There is no magic predictor.

The EasyLanguage and eSignal Formula Script (EFS) codes to compute several leading indicators are given in Figures 16.5 and 16.6, respectively. In these codes, the leading indicator is compared to an exponential moving average whose $\alpha = 0.5$. This exponential moving average has a lag of only a half bar. The relative positions of the leading indicator and the exponential moving average show when the market is in an uptrend or a downtrend as in the example in Figure 16.7. The alphas of the leading indicator are provided as inputs for ease of modification of the indicator. For example, the continuation of the trend is more clearly identified if $\alpha 1$ is reduced to a value of 0.15. The impact of giving the indicator greater lead is shown in Figure 16.8.

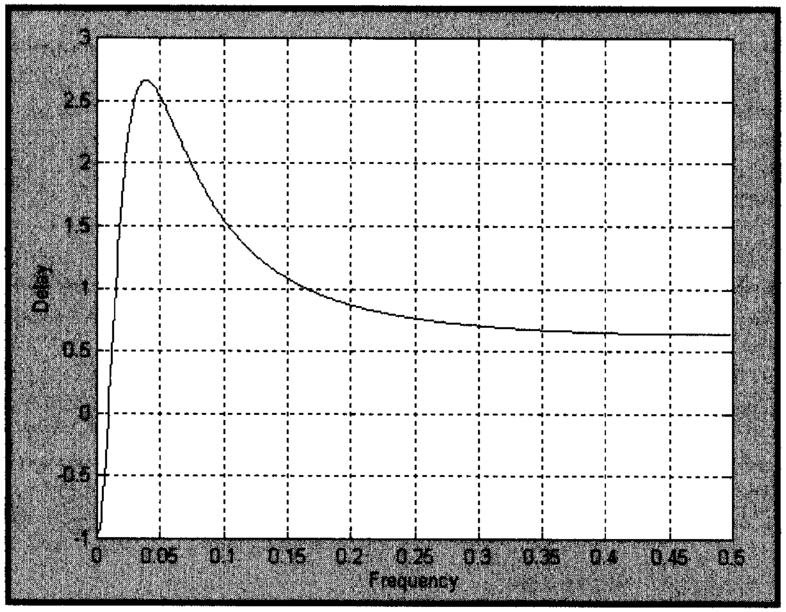

FIGURE 16.4 The Net Filter Has a Low-Frequency Leading Characteristic

```
Inputs: Price((H+L)/2),
        alpha1(.25),
        alpha2(.33);

Vars:   Lead(0),
        NetLead(0),
        EMA(0);

Lead = 2*Price +(alpha1 - 2)*Price[1]
   + (1 - alpha1)*Lead[1];
NetLead = alpha2*Lead + (1 - alpha2)*NetLead[1];

EMA = .5*Price + .5*EMA[1];

Plot1(NetLead, "Lead");
Plot2(EMA, "EMA");
```

FIGURE 16.5 EasyLanguage Code to Compute Leading Indicators

```
****************************************************
Title:  Leading Indicator
Coded By:  Chris D. Kryza (Divergence Software, Inc.)
Email:  c.kryza@gte.net
Incept:  09/02/2003
Version:  1.0.0


====================================================
Fix History:

09/02/2003 -   Initial Release
1.0.0

====================================================
***************************************************/


//External Variables
var nPrice                   = 0;
var nBarCount                = 0;

var aPriceArray              = new Array();
var aLead                    = new Array();
var aNetLead                 = new Array();
var aEMA                     = new Array();


//== PreMain function required by eSignal to set_
   things up
function preMain() {
var x;

  setPriceStudy(true);
  setStudyTitle("Leading Indicator");
  setCursorLabelName("Lead", 0);
  setCursorLabelName("EMA",  1 );
  setDefaultBarFgColor( Color.red,  0 );
  setDefaultBarFgColor( Color.blue, 1 );

      //initialize arrays
  for (x=0; x<10; x++) {
```

FIGURE 16.6 EFS Code to Compute Leading Indicators


```
        aPriceArray[x]             = 0.0;
        aLead[x]                   = 0.0;
        aNetLead[x]                = 0.0;
        aEMA[x]                    = 0.0;
  }

}

//== Main processing function
function main( Alpha1, Alpha2 ) {
var x;

       //initialize parameters if necessary
       if ( Alpha1 == null ) {
              Alpha1 = 0.25;
       }
       if ( Alpha2 == null ) {
              Alpha2 = 0.33;
       }

       // study is initializing
  if (getBarState() == BARSTATE_ALLBARS) {
    return null;
  }

       //on each new bar, save array values
       if ( getBarState() == BARSTATE_NEWBAR ) {

              nBarCount++;

              aPriceArray.pop();
              aPriceArray.unshift( 0 );

              aLead.pop();
              aLead.unshift( 0 );

              aNetLead.pop();
              aNetLead.unshift( 0 );

              aEMA.pop();
              aEMA.unshift( 0 );
```

(continued)

FIGURE 16.6 *(Continued)*

```
        }

        nPrice = ( high()+low() ) / 2;
        aPriceArray[0] = nPrice;


        aLead[0] = 2 * aPriceArray[0] + ( Alpha1 - 2.0 )
            * aPriceArray[1] + ( 1.0 - Alpha1 )
            * aLead[1];
        aNetLead[0] = Alpha2 * aLead[0]
            + ( 1.0 - Alpha2 ) * aNetLead[1];

        aEMA[0] = 0.5 * aPriceArray[0] + 0.5 * aEMA[1];


        //return the calculated values
        if ( !isNaN( aNetLead[0] ) && !isNaN( aEMA[0] )
            && nBarCount > 20 ) {
                return new Array( aNetLead[0], aEMA[0] );
        }


}
```

FIGURE 16.6 *(Continued)*

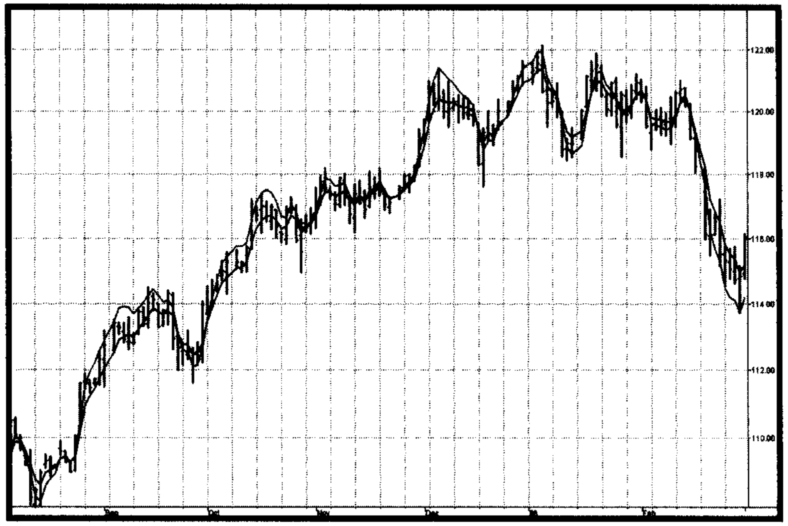

FIGURE 16.7 Leading Indicator ($\alpha 1 = 0.25$, $\alpha 2 = 0.33$) and EMA

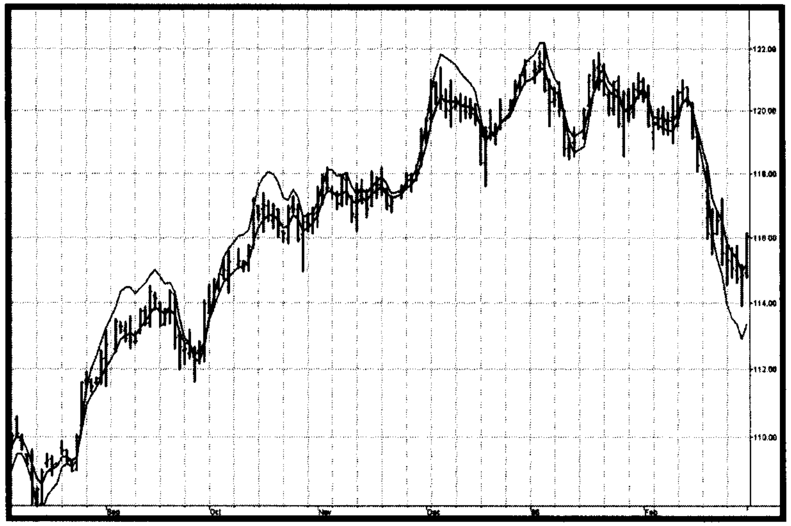

FIGURE 16.8 Leading Indicator ($\alpha 1 = 0.15$, $\alpha 2 = 0.33$) and EMA Provides a Clearer Picture of the Trend Continuation

KEY POINTS TO REMEMBER

- Adding the difference between price and an exponential moving average to the price itself creates a leading indicator.
- The leading indicator always has noise gain.
- Smoothing the leading indicator with another exponential moving average can mitigate noise gain.
- Constants can be selected to provide a net lead for the indicator at low frequencies.
- The leading indicator has a lagging signal at price turning points.

CHAPTER 17

Simplifying Simple Moving Average Computations

"One topic has to be last," said Tom finally.

A simple moving average (SMA) of length N is computed by adding N values and dividing the sum by N. The process is repeated on a bar-by-bar basis. What could be easier? While conceptually easy, the coding for long moving averages can be tedious because there are so many terms. The tedium can be reduced by putting the summation in a loop. But looping is difficult to do in some applications, such as Excel. Another simplifying approach is to drop off the oldest value and add a new value to the moving average. But this requires computing the initial value of the long moving average at least once. I will show you two ways to compute the SMA with ease.

In Z transform notation, a unit delay is represented by Z^{-1}. The transfer response is the output of the filter divided by its input. Thus, the transfer response of an eight-bar SMA would be written as

$$H(z) = (1 + Z^{-1} + Z^{-2} + Z^{-3} + Z^{-4} + Z^{-5} + Z^{-6} + Z^{-7})/8 \qquad (17.1)$$

This same expression, written in EasyLanguage where a delay of N bars is represented in square brackets as [N], is shown in Equation 17.2.

```
SMA = (Price + Price[1] + Price[2] + Price[3] + Price[3]
       + Price[4] + Price[5] + Price[6] + Price[7])/8;        (17.2)
```